

An economic model of reinforcement learning

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1 Definitions

Policy is a mapping from states to probabilities over action

$$\pi(a|s)$$

Q-function: State-action values of following a policy $\pi(a|s)$ in the future, and implement a specific a_t today

$$Q_\pi(s_t, a_t) = r(s_t, a_t) + \beta E_t Q_\pi(s_{t+1}, \pi(s_{t+1}))$$

Value function: value in state s of following policy $\pi(a|s)$

$$V_\pi(s) = \sum_a \pi(a|s) Q_\pi(s, a) \tag{1}$$

2 Bayesian resource-rational framework of learning

2.1 Object the agent is learning about

We assume the agent is trying to learn the true, full-information optimal Q-function $Q_{\pi^*}(s, a)$ where the optimal policy $\pi^*(a|s)$ satisfies the Bellman equation

$$V^*(s_t) = \max_{a_t} r(s_t, a_t) + \beta E_t (V(s_{t+1}))$$

The optimal policy function $\pi^*(a|s)$ defines the optimal Q-function:

$$Q_{\pi^*}(s_t, a_t) = r(s_t, a_t) + \beta E_t Q_{\pi^*}(s_{t+1}, \pi^*(s_{t+1}))$$

The agent will learn about the optimal Q-value function $Q_{\pi^*}(s, a)$ defined above.

2.2 Policies

Each period the agent enters with some beliefs about the Q-function, those beliefs follow a Gaussian Process distribution so that conditional on the $t - 1$ information set $\mathcal{I}_{t-1}$ (which is what the agent enters time t with)

$$Q_{\pi^*}(s, a)|\mathcal{I}_{t-1} \sim GP(\hat{Q}_{t-1}(s, a), \hat{\Sigma}_{t-1}(s, a, s', a'))$$

In period t , the agent follows the following policy function that defines his actions. Conditional on the (updated) information set $\mathcal{I}_t$, the probability of selecting action a given state s_t is

$$\pi_t(s_t) = \pi(\hat{Q}_t(s_t, a), \theta_t(s_t)) = \frac{\exp(\theta_t(s_t)\hat{Q}_t(s_t, a))}{\sum_{a'} \exp(\theta_t(s_t)\hat{Q}_t(s_t, a'))}$$

Here $\hat{Q}_t(s_t, a)$ is the conditional expectation of the Q-value function and $\theta_t(s_t)$ is a temperature parameter. We can treat the latter as a constant

$$\theta_t(s) = \theta, \text{ for all } (s, t)$$

or, more interestingly, have it take the following form:

$$\theta_t(s) = \theta_0 \frac{\Sigma_{V,0}(s)}{\Sigma_{V,t-1}(s)}$$

where $\Sigma_V(s)$ will be a measure of perceived uncertainty, which we will define below.

Also define the 'greedy' policy as

$$\begin{aligned} \pi_{t-1}^{greedy}(a_t = \hat{a}_{t-1}^{\max}(s_t)|s_t) &= 1 \\ \pi_{t-1}^{greedy}(a_t \neq \hat{a}_{t-1}^{\max}(s_t)|s_t) &= 0 \\ \hat{a}_{t-1}^{\max}(s_t) &= \arg \max_a \hat{Q}_{t-1}(s_t, a) \end{aligned}$$

2.3 Updating

For any state s , compute the expected return

$$\widehat{V}_{t-1}(s) = \sum_a \pi_{t-1}(a_t = a|s) \widehat{Q}_{t-1}(s, a)$$

This is a variable measurable in the time $t - 1$ information set, hence it is known at the beginning of time t . It is the expected value of the following random variable:

$$V_{t-1}^{true}(s) \equiv \sum_a \pi_{t-1}(a_t = a|s) Q_{\pi}^*(s, a)$$

The policy $\pi_{t-1}(a|s)$ is known as of time t , and the randomness comes from the uncertainty over the unknown optimal $Q_{\pi^*}(s, a)$.

Based on prior information, this random variable has a distribution

$$\begin{aligned} V_{t-1}^{true}(s_t) &\sim N(\widehat{V}_{t-1}(s_t), \Sigma_{V,t-1}(s_t)) \\ \Sigma_{V,t-1}(s_t) &= Var_{t-1}\left(\sum_a \pi_{t-1}(a|s_t) Q_{\pi^*}(s_t, a)\right) = \pi_{t-1}(a|s_t)' \widehat{\Sigma}_{t-1}(a, s_t, a', s_t) \pi_{t-1}(a'|s_t) \end{aligned}$$

where $\widehat{\Sigma}_{t-1}(a, s, a', s)$ is a two-dimensional matrix, which evaluates the variance-covariance of $Q_{\pi^*}(s, a)$ at a fixed s_t . So then $\pi_{t-1}(a|s_t)' \widehat{\Sigma}_{t-1}(a, s_t, a', s_t) \pi_{t-1}(a'|s_t)$ is the typical sandwich expression for the variance of the linear combination $\sum_a \pi_{t-1}(a|s_t) \widehat{Q}_{t-1}(s_t, a)$.

2.3.1 Deliberation (i.e. model-based reasoning)

Resource-rationality, conditional trigger of reasoning The agent looks at $\Sigma_{V,t-1}(s_t)$ and decides whether to engage in additional reasoning to reduce this uncertainty. The agent's objective function in deciding whether to engage in extra reasoning is a tracking problem of the form

$$\min_{\Sigma_{V,t}(s_t)} E \left(\widehat{V}_t(s_t) - V_{t-1}^{true}(s_t) \right)^2 + \kappa \ln \left(\frac{\Sigma_{V,t-1}(s_t)}{\Sigma_{V,t}(s_t)} \right)$$

where κ is the marginal cost of reducing further uncertainty through reasoning. This objective is then

$$\min_{\widetilde{\Sigma}_{V,t}(s_t)} \widetilde{\Sigma}_{V,t}(s_t) + \kappa \ln \left(\frac{\Sigma_{V,t-1}(s_t)}{\widetilde{\Sigma}_{V,t}(s_t)} \right).$$

where

$$\widetilde{\Sigma}_{V,t}(s_t) = Var\left(\sum_a \pi_{t-1}(a|s_t) Q_{\pi^*}(s, a) | \mathcal{I}_{t-1}, \eta_t^R\right)$$

where $Var(Q_{\pi^*}(s, a)|\mathcal{I}_{t-1}, \eta_t^R)$ is the updated conditional uncertainty after observing the new signal η_t^R , which we define below. The optimal information choice is

$$\tilde{\Sigma}_{V,t} = \min\{\kappa, \Sigma_{V,t-1}(s_t)\} \quad (2)$$

2.3.2 Mental simulation, model-based learning

Each period t , internal deliberation produces a noisy signal $\eta_t^R(s_t, a_t)$

$$\eta_t^R = \sum_a \pi_{t-1}(a|s_t) Q_{\pi^*}(s_t, a) + \varepsilon_t^R$$

where $\varepsilon_t^R \sim N(0, \sigma_{\eta^R,t}^2)$, where the variance of the noise follows from the above optimal information choice in (2):

$$\sigma_{\eta^R,t}^2 = \begin{cases} \frac{\kappa \Sigma_{V,t-1}(s_t)}{\Sigma_{V,t-1}(s_t) - \kappa} & , \text{ if } \Sigma_{V,t-1}(s_t) > \kappa. \\ \infty & , \text{ if } \Sigma_{V,t-1}(s_t) \leq \kappa. \end{cases}$$

Then use that signal to update beliefs

$$\hat{Q}_t^R(s, a) = \hat{Q}_{t-1}(s, a) + \alpha_t^R(s, a) \left(\eta_t^R - \sum_{a'} \pi_{t-1}(a'|s_t) \hat{Q}_{t-1}(s_t, a') \right)$$

where we can derive $\alpha_t^R(s, a)$ as follows:

$$\begin{aligned} \alpha_t^R(s, a) &= \frac{Cov_{t-1}(Q_{\pi^*}(s, a), \sum_{a'} \pi_{t-1}(a'|s_t) Q_{\pi^*}(s_t, a'))}{Var_{t-1}(\sum_{a'} \pi_{t-1}(a'|s_t) Q_{\pi^*}(s_t, a') + \varepsilon_t^R)} \\ &= \sum_{a'} \pi_{t-1}(a'|s_t) \frac{Cov(Q_{\pi^*}(s, a), Q_{\pi^*}(s_t, a'))}{\Sigma_{V,t-1}(s_t) + \sigma_{\eta^R,t}^2} \end{aligned}$$

Similarly, posterior variances and etc. of $Q_{\pi^*}(s, a)$ update as well at all (s, a) pairs.

Cost of reasoning is the reduction of uncertainty (at optimal reasoning)

$$c_{t-1/t}(s_t) \equiv \kappa \ln \left(\frac{\Sigma_{V,t-1}(s_t)}{\Sigma_{V,t}(s_t)} \right)$$

By optimal reasoning this is only a function of previous uncertainty, evaluated at current state

$$c_{t-1}(s_t) = \begin{cases} \kappa \ln\left(\frac{\Sigma_{V,t-1}(s_t)}{\kappa}\right) & , \text{ if } \Sigma_{V,t-1}(s_t) > \kappa. \\ 0 & , \text{ if } \Sigma_{V,t-1}(s_t) \leq \kappa. \end{cases}$$

Therefore experienced reward at any time t is

$$r(s_t, a_t) \equiv u(s_t, a_t) - c(s_t)$$

where $u(s_t, a_t)$ might be physical utility.

2.3.3 Model-free learning

Each period the agent also observes the utility he just experienced at the end of last period, $u(s_{t-1}, a_{t-1})$. He can use that observed flow utility value to update beliefs about $Q_{\pi^*}(s, a)$ as well. We think of this learning “by experience” as free, so it will not be subject to a cognitive cost. Still, it is limited as we can see below.

It is most intuitive to write this “experience” signal as

$$\eta_t^Q = r(s_{t-1}, a_{t-1}) + \beta \sum_a \sum_{s_t} \pi_{t-1}^{greedy}(a|s_t) \hat{Q}_{t-1}(s_t, a) F(s_t|s_{t-1})$$

where $F(s_t|s_{t-1})$ is the law of motion for the state (that could also depend on the choice a_{t-1} , but for simplicity we suppress this notation). So the signal η_t^Q is sort of like the expected value of $Q(s_{t-1}, a_{t-1})$ from the perspective of time $t - 1$. Note: we use the greedy policy because we’re learning about the (unknown) Q-value derived under the (unknown) greedy policy.

The new “information” is in the fact that we will now impose the Q-value equation

$$Q_{\pi^*}(s_t, a_t) = r(s_t, a_t) + \beta E_t Q_{\pi^*}(s_{t+1}, \pi^*(s_{t+1}))$$

which up to this point the agent has never used directly (only abstractly, in reduced form in the noisy signals $\eta_{\pi^R, t}$. Important: we do not assume this equation is in the agent’s information set. Assuming knowledge of this equation automatically means we are assuming the agent knows $Q_{\pi^*}(s, a)$, since the optimal value function is defined by that equation and is just a direct one-to-one map from it.

Note: we can just rewrite that as something that makes explicit the greedy policy

$$Q_{\pi^*}(s_t, a_t) = r(s_t, a_t) + \beta E_t \max_{a'} Q_{\pi^*}(s_{t+1}, a')$$

So we are not going to give the agent full knowledge of this equation, we'll just let it use once in a way. Specifically, we will use it to express the experience signal as

$$\begin{aligned}\eta_t^Q &= u(s_{t-1}, a_{t-1}) = r(s_{t-1}, a_{t-1}) + \beta \sum_a \sum_{s_t} \pi_{t-1}^{greedy}(a|s_t) \hat{Q}_{t-1}(s_t, a) F(s_t|s_{t-1}) \\ &= Q^*(s_{t-1}, a_{t-1}) + \beta \sum_a \sum_{s_t} \pi_{t-1}^{greedy}(a|s_t) F(s_t|s_{t-1}) \left(\hat{Q}_{t-1}(s_t, a) - Q_{\pi^*}(s_t, a) \right)\end{aligned}$$

The agent uses that to update beliefs

$$\hat{Q}_t(s, a) = \hat{Q}_{t-1}(s, a) + \alpha_t^Q(s, a) \left[\eta_t^Q - \hat{Q}_{t-1}(s_{t-1}, a_{t-1}) \right]$$

where

$$\alpha_t^Q(s, a) = \frac{Cov_{t-1}(\eta_t^Q, Q^*(s, a))}{Var_{t-1}(\eta_t^Q)}$$

with

$$\begin{aligned}Cov_{t-1}(\eta_t^Q, Q^*(s, a)) &= Cov_{t-1}(Q^*(s_{t-1}, a_{t-1}) - \beta \sum_a \sum_{s_t} \pi_{t-1}^{greedy}(a|s_t) F(s_t|s_{t-1}) Q^*(s_t, a_t), Q^*(s, a)) \\ &= Cov_{t-1}(Q^*(s_{t-1}, a_{t-1}), Q^*(s, a)) - \beta \sum_a \sum_{s_t} \pi_{t-1}^{greedy}(a|s_t) F(s_t|s_{t-1}) Cov_{t-1}(Q^*(s_t, a_t), Q^*(s, a))\end{aligned}$$

The key is that at this step, when we compute the signal-to-noise ratio, we DO NOT impose the Q-value equation *again*, because this would imply

$$Q^*(s_{t-1}, a_{t-1}) - \beta \sum_a \sum_{s_t} \pi_{t-1}^{greedy}(a|s_t) f(s_t|s_{t-1}) Q^*(s_t, a_t) = r(s_{t-1}, a_{t-1}),$$

which is a known constant in the information set. By not imposing knowledge of that equation again, and computing the signal-to-noise ratio by simply evaluating the above covariances under the $t - 1$ information set as recursively defined is like doing only a one step update based on the Qvalue equation, but not an infinite update. And this makes sense – infinitely iterating forward on the Bellman equation is solving it. We want to impose a cost on solving it, so any deliberate iteration and attempts to solve the Bellman equation only come through the costly signals η_t^R .

3 Joint actions and reasoning choices

Conditional on a realized s_t , the objective function of the agent is to choose an action policy $\pi(a|s_t)$ and reasoning choices over signals η_t , resulting in posterior variance of beliefs $\tilde{\Sigma}_t$, to maximize

$$\max_{\pi(a|s_t), \Sigma_t} \sum_a \hat{Q}_t(a, s_t) \pi(a|s_t) - w \sum_a \tilde{\sigma}_t^2(a, s_t) - \frac{\kappa}{2} \ln \left[\frac{|\Sigma_{t-1}(a, a'; s_t)|}{|\tilde{\Sigma}_t(a, a'; s_t)|} \right]$$

subject to two constraints

$$\begin{aligned} \sum_a \pi(a|s_t) &= 1 \\ \underbrace{- \sum_a \ln(\pi(a|s_t)) \pi(a|s_t)}_{\equiv H_\pi(s_t)} &\geq h \sum_a \tilde{\sigma}_t^2(a, s_t) \end{aligned}$$

where

- $\Sigma_{t-1}(a, a'; s_t) = \text{Var}(Q^*(a, s_t) | \mathcal{I}_{t-1})$ is the beginning of period t variance that agents perceive over the vector $Q^*(a, s_t)$ at a given s_t (it is a vector due to the vector of possible actions a), which variance is conditional on the end of period $t-1$ information set $\mathcal{I}_{t-1}$.
- $\tilde{\Sigma}_t(a, a'; s_t) = \Sigma_{t-1}(a, a'; s_t | \eta_t) = \text{Var}(Q^*(a, s_t) | \mathcal{I}_{t-1}, \eta_t)$ is the posterior variance conditional on the time $t-1$ information set $\mathcal{I}_{t-1}$ and also the new signals η_t . Note, this is not the end-of-period t posterior variance $\Sigma_t(a, a'; s_t)$ because at the end of the period you also receive and update based on the experience signal (discussed below).
- $\tilde{\sigma}_t^2(a, s)$ is the diagonal entry of $\Sigma_{t-1}(a, a'; s_t | \eta_t)$ corresponding to the position of action a .
- $H_\pi(s_t) \equiv - \sum_a \ln(\pi(a|s_t)) \pi(a|s_t)$ is entropy of the actions distribution, conditional on a state s_t .

This objective function captures:

- benefit of exploitation (greediness): incentive to choose the action with highest expected value, through the linear term $\sum_a \hat{Q}_t(a, s) \pi(a|s)$
- possible disutility cost of uncertainty over the values associated to each action (the term $w \sum_a \tilde{\sigma}_t^2(a, s_t)$). Essentially, a direct cost of cognitive dissonance, when $w > 0$. This component going through w allows us to have a benefit of reasoning that is separate from the exploitation-experimentation tradeoff below.

- constant marginal cost of cognitive reasoning, when $\kappa > 0$. Note:
 - we do not model reasoning as a fixed capacity, but as a continuous choice
 - The term $\frac{1}{2} \ln \left[\frac{|\Sigma_{t-1}(a, a'; s_t)|}{|\tilde{\Sigma}_t(a, a'; s_t)|} \right]$ in the objective function is the information content of the signals η_t , measured in entropy – i.e. that is the reduction in entropy after observing the vector of signals η_t .
- The first constraint essentially constraints $\pi(a|s_t)$ to be proper probability distribution (it will be associated with a Lagrange multiplier $\lambda > 0$)
- The second constraint is an entropy regularization and its aim is to capture the desire for exploration/experimentation. That is, as long as the value function $Q^*(s, a)$ is uncertainty, there is a desire to experiment and explore the function in other parts of the state space (exploration is valuable here because of the model free signal agents get each period, for free). We model this desire to experiment along the idea of “entropy regularization” from ML literature, which essentially requires the entropy of the action distribution to be bigger than some minimum threshold, thus ensuring randomization. Nevertheless, we recognize that there is only reason to experiment to the extent to which the value function is uncertainty – if $Q^*(a, s_t)$ was known, then there is no point to experimenting as there is nothing to learn. Hence, we make the lower bound on the entropy of the action distribution $H_\pi(s_t)$ proportional to the remaining uncertainty over $Q^*(a, s_t)$, measured by $\sum_a \sigma_t^2(a, s_t)$.

With the LMs, objective is

$$\begin{aligned} & \max_{\pi_t(a|s_t), \tilde{\Sigma}_t} \sum_a \hat{Q}_t(a, s_t) \pi(a|s_t) - w \sum_a \tilde{\sigma}_t^2(a, s_t) - \frac{\kappa}{2} \ln \left[\frac{|\Sigma_{t-1}(a, a'; s_t)|}{|\tilde{\Sigma}_t(a, a'; s_t)|} \right] \\ & + \delta_t \left[-h \sum_a \tilde{\sigma}_t^2(a, s_t) - \sum_a \ln(\pi(a|s_t)) \pi(a|s_t) \right] + \lambda_t (\sum_a \pi(a|s) - 1) \end{aligned}$$

- **Optimal actions.** The optimal policy $\pi(a|s_t)$ looks like a softmax, with a state-dependent temperature

$$\hat{\pi}(a|s_t) = \begin{cases} \frac{\exp(\frac{\hat{Q}_t(a, s_t)}{\delta_t})}{\sum_a \exp(\frac{\hat{Q}_t(a, s_t)}{\delta_t})} & \text{if } \delta_t > 0 \\ \hat{\pi}^{greedy}(a|s_t) & \text{if } \delta_t = 0 \end{cases}$$

where $\hat{\pi}^{greedy}(a|s_t)$ is the degenerate policy with probability 1 on $\arg \max Q(s_t, a)$.

To finding $\delta_t > 0$, plug in this expression into the constraint and solve given the optimal info choice (which we characterize below)

$$\sum_a \ln\left(\frac{\exp(\frac{\hat{Q}_t(a,s)}{\delta_t})}{\sum_a \exp(\frac{\hat{Q}_t(a,s)}{\delta_t})}\right) \frac{\exp(\frac{\hat{Q}_t(a,s)}{\delta_t})}{\sum_a \exp(\frac{\hat{Q}_t(a,s)}{\delta_t})} = h \sum_a \tilde{\sigma}_t^2(a, s)$$

Limit under $\kappa = 0$. To get some intuition, it is useful to consider the limit of costless reasoning $\kappa = 0$. Formally, the objective becomes

$$\max_{\pi_t(a|s_t), \Sigma_t} \sum_a \hat{Q}_t(a, s_t) \pi(a|s_t) - w \sum_a \tilde{\sigma}_t^2(a, s_t) + \delta_t \left[-h \sum_a \tilde{\sigma}_t^2(a, s_t) - \sum_a \ln(\pi(a|s_t)) \pi(a|s_t) \right] + \lambda_t (\sum_a \pi(a|s_t) - 1)$$

Given costless reasoning, we will have $\tilde{\sigma}_t^2(a, s_t) = 0$. In this case, the entropy regularization constraint will not bind. So $\delta_t = 0$ and the optimal action follows the usual greedy policy. Further note that $\tilde{\sigma}_t^2(a, s_t) = 0$ means the agent knows the $Q(a, s_t)$ associated to each action, conditional on current state s_t and conditional on following the optimal policy from now on. Thus in this limit agent knows $Q_{\pi^*}(a, s)$ and we can replace

$$\hat{Q}_t(a, s) = Q_{\pi^*}(a, s)$$

Objective then is

$$\max_{\pi_t(a|s_t)} \sum_a Q_{\pi^*}(a, s_t) \pi(a|s_t) + \lambda_t (\sum_a \pi(a|s_t) - 1)$$

which is exactly the problem solved under full rationality in the first place:

$$\begin{aligned} \max_{a_t} Q_{\pi^*}(a_t, s_t) &= \max_{a_t} \left[u(a_t, s_t) + \beta E_t \max_{a'} Q_{\pi^*}(a', s_{t+1}) \right] \\ \pi^*(a|s) &= \pi^{greedy}(a|s) \end{aligned}$$

So, in the limit where the agent learns fully, we are back to the full information problem as expected.

- **Reasoning choice.** The agent learns about the full-information $Q_{\pi^*}(a, s)$. The agent has can generate reasoning signals η_t that are unrestricted linear functions of the underlying $Q_{\pi^*}(a, s_t)$

$$\eta_t = V_t' Q_{\pi^*}(a, s_t) + \varepsilon_{\eta,t}$$

where $\varepsilon_{\eta,t} \sim N(0, \Sigma_{\eta,t})$. The agents freely decide on the structure of V_t and the noise variance matrix $\Sigma_{\eta,t}$, with only imposing the normalization that the columns of V_t have

a norm of 1.

Using the following two properties of eigenvalues,

$$\begin{aligned} \text{tr}(\Sigma_t(a, a'; s_t)) &= \sum_i \lambda_t^{(i)} \\ |\Sigma_t(a, a'; s_t)| &= \prod_i \lambda_t^{(i)} \end{aligned}$$

where $\lambda_t^{(i)}$ are the eigenvalues of $\tilde{\Sigma}_t(a, a'; s_t)$, we can rewrite the objective function as

$$\begin{aligned} \max_{\pi_t(a|s_t), \Sigma_t} \sum_a \hat{Q}_t(a, s_t) \pi(a|s_t) - w \sum_i \lambda_t^{(i)} - \frac{\kappa}{2} \ln \left[\ln(|\Sigma_{t-1}(a, a'; s_t)|) - \sum_i \ln(\lambda_t^{(i)}) \right] \\ + \delta_t \left[-h \sum_i \lambda_t^{(i)} - \sum_a \ln(\pi(a|s_t)) \pi(a|s_t) \right] + \lambda_t \left(\sum_a \pi(a|s) - 1 \right) \end{aligned}$$

The FOCs are

$$\lambda_t^{(i)} = \frac{\kappa}{w + \delta_t h}$$

Thus, we can see the interaction between actions and reasoning – there is an incentive to reason more when the LM δ_t is larger, when you would value relaxing the constraint on greediness induced by experimentation. The agent can relax that constraint by reducing uncertainty. Intuitively, the agent values the reduction in uncertainty because that allows it to exploit $\hat{Q}_t(a, s_t)$ more, in the sense of selecting an action closer to its argmax.

Also, the above manipulations show that the solution to δ_t should satisfy

$$\sum_a \ln \left(\frac{\exp(\frac{\hat{Q}_t(a, s)}{\hat{\delta}_t})}{\sum_a \exp(\frac{\hat{Q}_t(a, s)}{\hat{\delta}_t})} \right) \frac{\exp(\frac{\hat{Q}_t(a, s)}{\hat{\delta}_t})}{\sum_a \exp(\frac{\hat{Q}_t(a, s)}{\hat{\delta}_t})} = h \sum_i \lambda_t^{(i)}$$

Without loss of generality, we can assume that $\lambda_t^{(i)}$ are sorted in descending order so

$$\lambda_t^{(1)} \geq \lambda_t^{(2)} \geq \dots \geq \lambda_t^{(N)}$$

Importantly, by standard properties of variance matrices and conditional variances, since Σ_η is positive definite by assumption, the (sorted) eigenvalues of $\Sigma_t(a, a'; s_t)$ must

be weakly smaller than those of the prior variance $\Sigma_{t-1}(a, a'; s_t)$. Essentially, this is the “no forgetting” constraint from our univariate case, and it implies that

$$\lambda_t^{(i)} \leq \lambda_{t-1}^{(i)}$$

So imposing the no-forgetting constraint we see that the optimal information choice over the structure and variances of the signals η_t must be such that

$$\lambda_t^{(i)} = \min\left\{\frac{\kappa}{w + \delta_t h}, \lambda_{t-1}^{(i)}\right\}$$

Hence, agents want to only reduce eigenvalues that are already bigger than κ . But any information flow that reduces any of the eigenvalues that already below κ is inefficient. Hence, the optimal signal structure is for V_t to be the matrix of eigenvectors of the prior variance $\Sigma_{t-1}(a, a'; s_t)$, and $\Sigma_{\eta,t}$ to be a diagonal matrix with diagonal entries $\sigma_{i,t}^2$ where

$$\sigma_{i,t}^2 = \begin{cases} \frac{\kappa \lambda_{t-1}^{(i)}}{\lambda_{t-1}^{(i)} - \kappa}, & \text{if } \lambda_t^{(i)} > \kappa \\ \infty, & \text{if } \lambda_t^{(i)} \leq \kappa \end{cases}$$

Typically, the eigenvalues are declining in value quickly, so only a few of these signals will have non-infinite noise variance. Moreover the update from $\lambda_{t-1}^{(i)}$ to $\lambda_t^{(i)}$ is straightforward because it is an update over orthogonal normal variables with a vector of orthogonal signals.

End of period t – Model Free learning. The above joint solution produces optimal reasoning signals η_t , and after observing those signals and following the optimal policy function $\hat{\pi}(a|s_t)$, also a realized action a_t . Given that action, at the end of the period the agent observes realized utility $u(s_t, a_t)$. This knowledge implies an “experience” or model-free learning signal

$$\begin{aligned} \eta_t^Q &= u(s_t, a_t) + \beta \sum_a \sum_{s_{t+1}} \pi_t^{\text{greedy}}(a|s_{t+1}) \hat{Q}_t(s_{t+1}, a) F(s_{t+1}|s_t) \\ &= u(s_t, a_t) + \beta \sum_a \sum_{s_{t+1}} \left[\pi_t^{\text{greedy}}(a|s_{t+1}) \hat{Q}_t(s_{t+1}, a) + \pi^*(a|s_{t+1}) Q_{\pi^*}(s_{t+1}, a) - \pi^*(a|s_{t+1}) Q_{\pi^*}(s_{t+1}, a) \right] F(s_{t+1}|s_t) \\ &= Q^*(s_t, a_t) + \beta \sum_a \sum_{s_{t+1}} \left[\pi_t^{\text{greedy}}(a|s_{t+1}) \hat{Q}_t(s_{t+1}, a) - \pi^*(a|s_{t+1}) Q_{\pi^*}(s_{t+1}, a) \right] F(s_{t+1}|s_t) \end{aligned}$$

where $\pi_t^{greedy}(a|s_{t+1}) = \arg \max_a \widehat{Q}_t(s_{t+1}, a)$. Because the greedy policies are degenerate, the last line simplifies to

$$\eta_t^Q = Q^*(s_t, a_t) + \beta \sum_{s_{t+1}} \left[\widehat{Q}_t(s_{t+1}, \pi_t^{greedy}(a|s_{t+1})) - Q_{\pi^*}(s_{t+1}, \pi^*(a|s_{t+1})) \right] F(s_{t+1}|s_t)$$

The agent has beliefs about $Q_{\pi^*}(s_t, a)$ for all actions a but does not know $\pi^*(a|s_{t+1})$. To start with, let us approximate by setting $\pi^*(a|s_{t+1}) = \pi_t^{greedy}(a|s_{t+1})$

$$\eta_t^Q = Q^*(s_t, a_t) + \beta \sum_{s_{t+1}} \left[\widehat{Q}_t(s_{t+1}, \pi_t^{greedy}(a|s_{t+1})) - Q_{\pi^*}(s_{t+1}, \pi_t^{greedy}(a|s_{t+1})) \right] F(s_{t+1}|s_t)$$

and use this signal to update beliefs about $Q(s, a)$ as per formulas above. Note: the approximation $\pi^*(a|s_t) = \pi_{t-1}^{greedy}(a|s_t)$ introduces another layer of noise, which we could potentially add in the future, but for now let's ignore it.

Using this signal, we update the beliefs of the agent one last time to obtain the end of period t beliefs, after both active deliberation and experiential learning.

4 Application: firms' investment

Consider a firm j 's investment problem. Starts the period with some capital $k_{j,t}$. Budget constraint says dividends equal

$$d_{j,t} = z_{j,t} k_{j,t}^\alpha + i_{j,t} [1 - \phi(i_{j,t}, k_{j,t})]$$

where $z_{j,t} k_{j,t}^\alpha$ is output of firm, $z_{j,t}$ is a technology shock, $i_{j,t}$ is investment and $\phi(i_{j,t}, k_{j,t})$ could be some investment adjustment costs. For example, it might be the usual i/k form:

$$\phi(i_{j,t}, k_{j,t}) = \varkappa \frac{i_{j,t}}{k_{j,t}}$$

Capital accumulation:

$$k_{j,t+1} = (1 - \delta)k_{j,t} + i_{j,t}$$

and a law of motion for technology $F(z_{j,t+1}|z_{j,t})$, for example an AR(1).

Firm's problem is

$$V(k_{j,t}, z_{j,t}) = \max_{i_{j,t}} [u(d_{j,t}) + \beta E_t V(k_{j,t+1}, z_{j,t+1})]$$

where $u(d_{j,t})$ is current payoff function, over dividends. We can have linear or concave $u(\cdot)$,

for example $u(d) = \log(d)$.

For now, we make entry and exit completely exogenous. With some probability γ , an individual firm j exits the economy. In particular, a new firm replaces it. The new firm will have some time-zero prior beliefs over the objects defined in previous sections (and further below), and have some starting capital $k_{j,0} = k_0$. We will endogenize at some point more the entry and exit decisions.

Because all firms will solve separately their problem there is no need to keep track of the j index for what follows.

Call the firm's state $s_t = (k_t, z_t)$. We would then need to:

1. solve for the fully rational policy function $i^*(s_t)$.

2. Implement our learning model, as described above. Recall the optimal Q-function, where the action is now investment

$$Q_{i^*}(s_t, i_t) = r(s_t, i_t) + \beta E_t Q_{i^*}(s_{t+1}, i^*(s_{t+1}))$$

where $r(s_t, i_t) = u(z_t k_t^\alpha + i_t - \phi(i_t, k_t))$. The agent will learn about the optimal Q-value function $Q_{\pi^*}(s, i)$ defined above.

To start with we could even make it simpler, but shutting down stochastic shock z_t . Instead it's entirely about transitional dynamics, through the endogenous state k_t .